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Set 11

Product Category Analysis

Dataset: Product Categories

Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

11 a. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
library(ggplot2)
```

```
data <- data.frame(
```

```
Category=c("Electronics","Clothing","Appliances"),
```

```
Sales=c(50000,35000,40000)
```

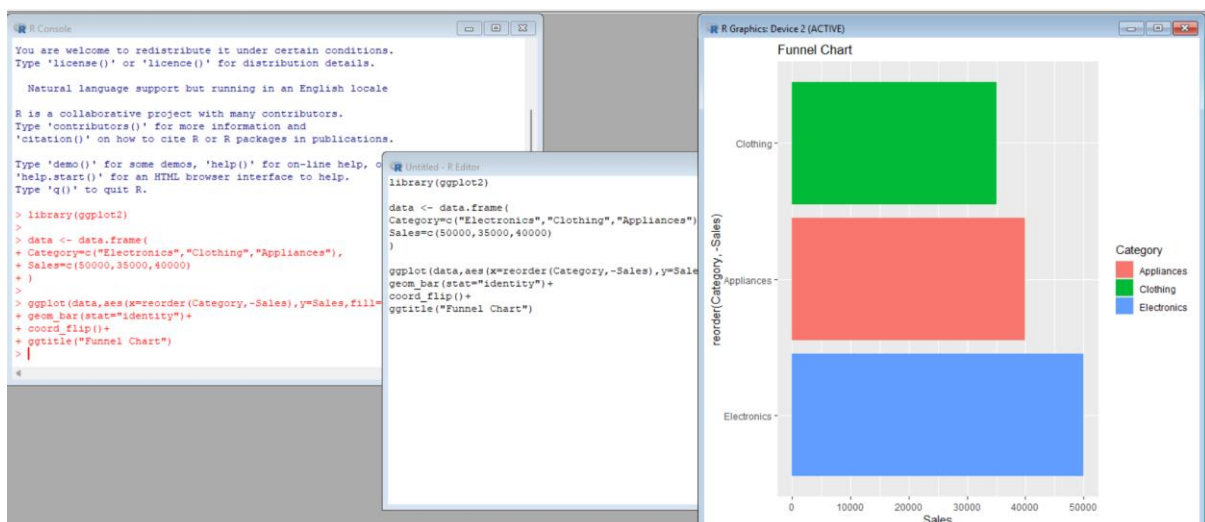
```
)
```

```
ggplot(data,aes(x=reorder(Category,-Sales),y=Sales,fill=Category))+
```

```
geom_bar(stat="identity")+
```

```
coord_flip()+
```

```
ggtitle("Funnel Chart")
```



11b. Build a table to display the sales data for each product category. Label the table elements.

```
data <- data.frame(
  Category=c("Electronics","Clothing","Appliances"),
  Sales=c(50000,35000,40000)
)
print(data)
```

```
  Category Sales
1 Electronics 50000
2  Clothing  35000
3 Appliances 40000
```

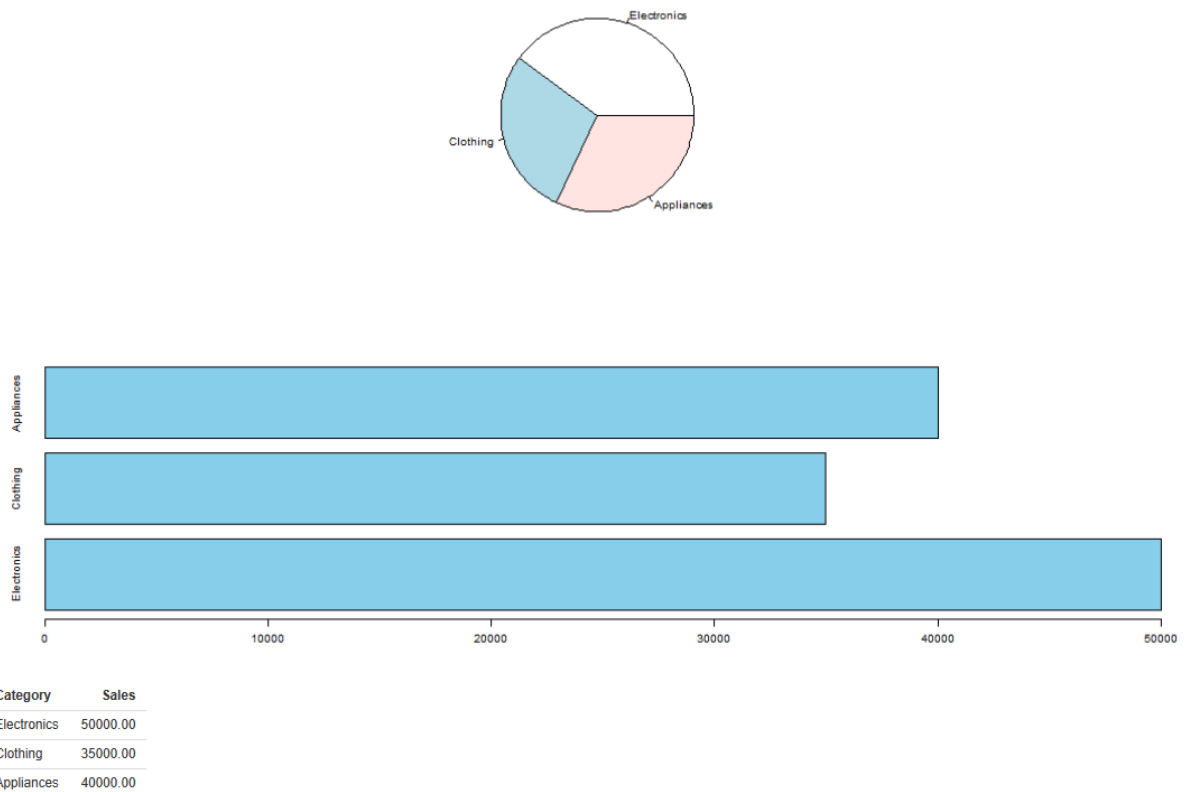
11c. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

```
library(shiny)
library(ggplot2)
data <- data.frame(
  Category=c("Electronics","Clothing","Appliances"),
  Sales=c(50000,35000,40000)
)
ui<-fluidPage(
  plotOutput("pie"),
  plotOutput("funnel"),
  tableOutput("table")
)
server<-function(input,output){
  output$pie<-renderPlot({
    pie(data$Sales,labels=data$Category)
  })
  output$funnel<-renderPlot({
    barplot(data$Sales,names.arg=data$Category,horiz=TRUE,col="skyblue")
  })
  output$table<-renderTable({
    data
```

```

})
}
shinyApp(ui,server)

```

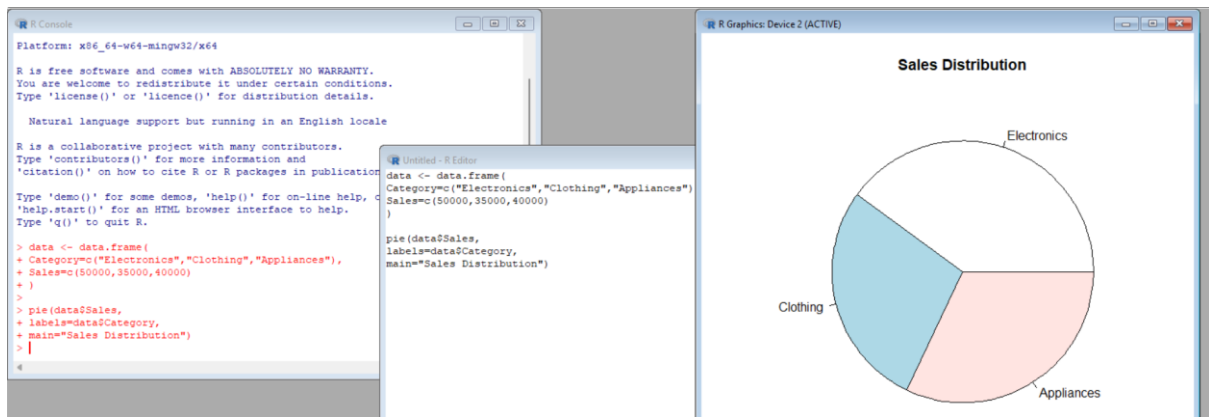


11d. Create a pie chart to represent the distribution of sales across product categories. Include labels.

```

data <- data.frame(
  Category=c("Electronics","Clothing","Appliances"),
  Sales=c(50000,35000,40000)
)
pie(data$Sales,
  labels=data$Category,
  main="Sales Distribution")

```

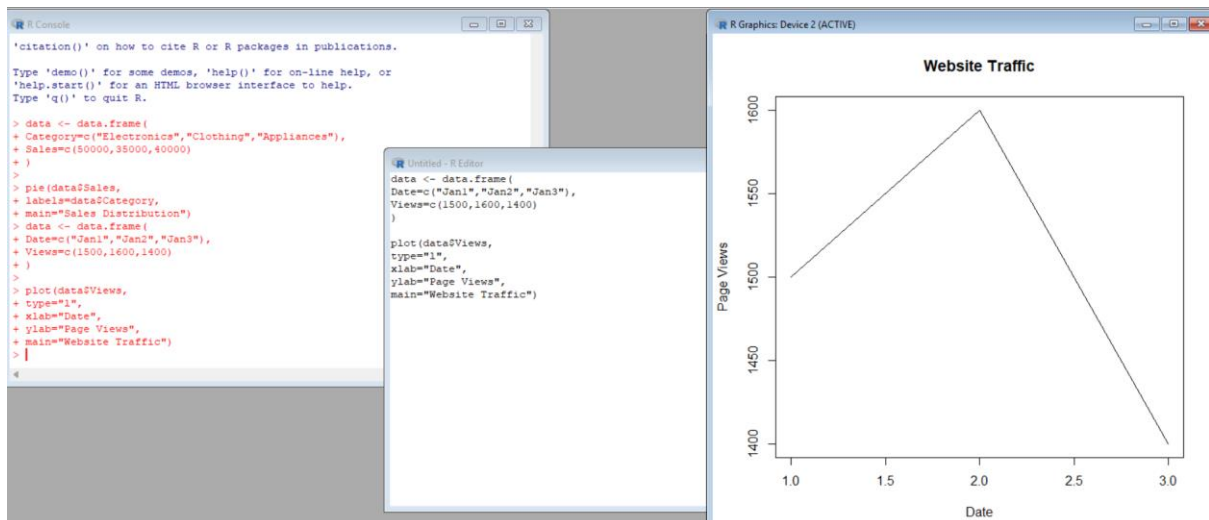


Set 12 :Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

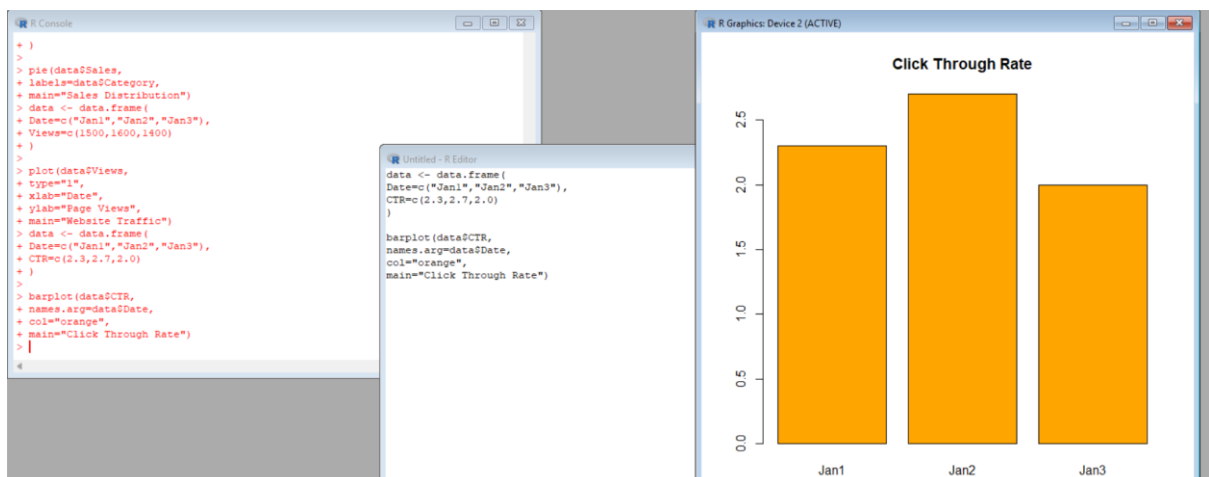
12a. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
data <- data.frame(
  Date=c("Jan1","Jan2","Jan3"),
  Views=c(1500,1600,1400)
)
plot(data$Views,
  type="l",
  xlab="Date",
  ylab="Page Views",
  main="Website Traffic")
```



12b. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
data <- data.frame(
  Date=c("Jan1","Jan2","Jan3"),
  CTR=c(2.3,2.7,2.0)
)
barplot(data$CTR,
  names.arg=data$Date,
  col="orange",
  main="Click Through Rate")
```



12c. Build a table to show the website traffic data for each day. Label the table elements.

```
data <- data.frame(
  Date=c("2023-01-01","2023-01-02","2023-01-03"),
  Views=c(1500,1600,1400),
```

```
CTR=c("2.3%","2.7%","2.0%")
```

```
)
```

```
print(data)
```

```
      Date Views  CTR
1 2023-01-01  1500 2.3%
2 2023-01-02  1600 2.7%
3 2023-01-03  1400 2.0%
```

12d. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

```
library(shiny)
```

```
ui<-fluidPage(
```

```
  plotOutput("line"),
```

```
  plotOutput("bar"),
```

```
  tableOutput("table")
```

```
)
```

```
server<-function(input,output){
```

```
  data<-data.frame(
```

```
    Date=c("Jan1","Jan2","Jan3"),
```

```
    Views=c(1500,1600,1400),
```

```
    CTR=c(2.3,2.7,2.0)
```

```
  )
```

```
  output$line<-renderPlot({
```

```
    plot(data$Views,type="l")
```

```
  })
```

```
  output$bar<-renderPlot({
```

```
    barplot(data$CTR,names.arg=data$Date)
```

```
  })
```

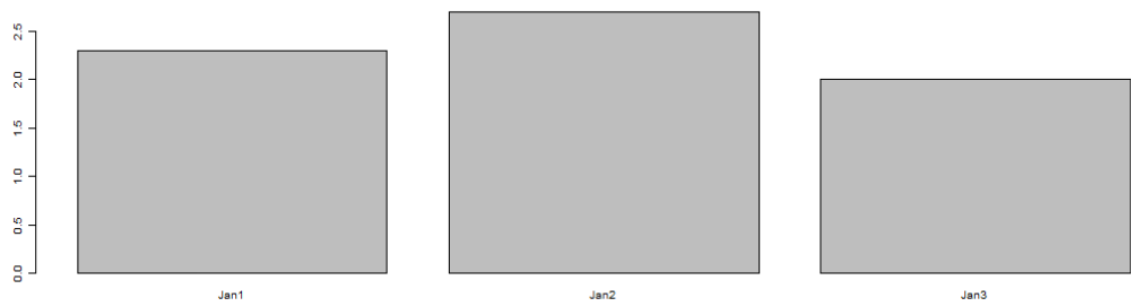
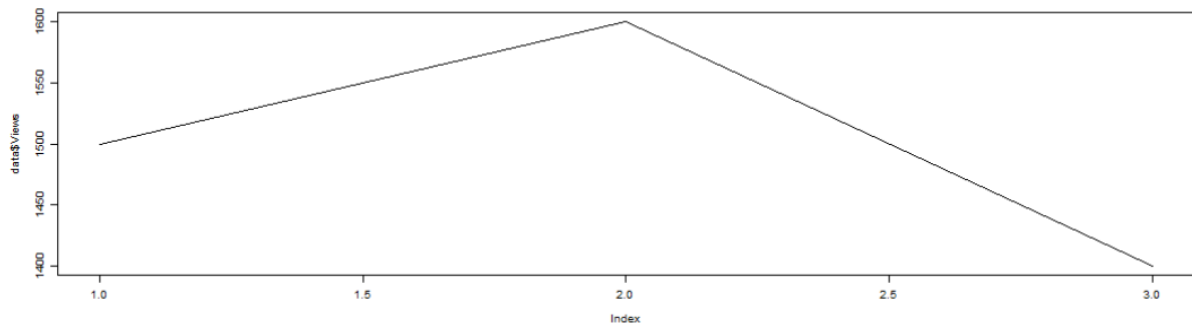
```
  output$table<-renderTable({
```

```
    data
```

```
  })
```

```
}
```

```
shinyApp(ui,server)
```



Date	Views	CTR
Jan1	1500.00	2.30
Jan2	1600.00	2.70
Jan3	1400.00	2.00

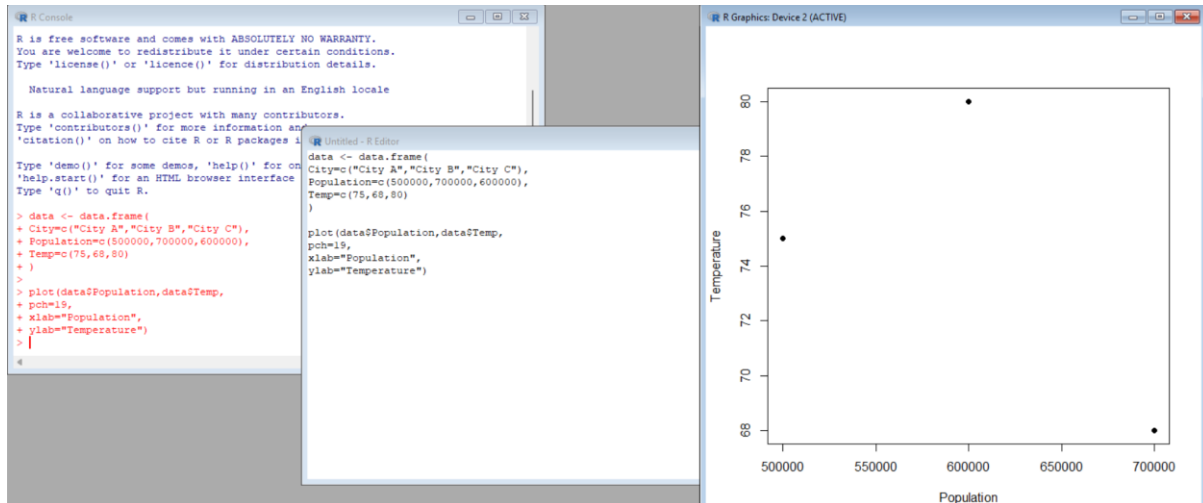
13. Dataset: Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

13a. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

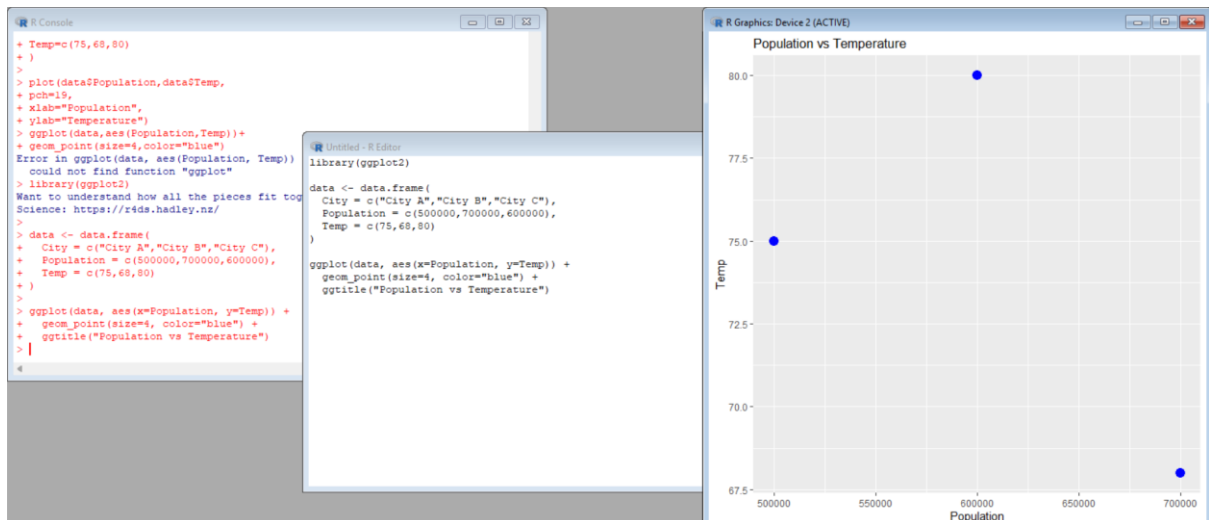
```
data <- data.frame(
  City=c("City A","City B","City C"),
  Population=c(500000,700000,600000),
  Temp=c(75,68,80)
)
```

```
plot(data$Population,data$Temp,
pch=19,
xlab="Population",
ylab="Temperature")
```



13b. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

```
library(ggplot2)
data <- data.frame(
  City = c("City A","City B","City C"),
  Population = c(500000,700000,600000),
  Temp = c(75,68,80)
)
ggplot(data, aes(x=Population, y=Temp)) +
  geom_point(size=4, color="blue") +
  ggtitle("Population vs Temperature")
```

13c. Build a table to display the geographic data for each city. Label the table elements.

```

data <- data.frame(
  City=c("City A","City B","City C"),
  Population=c(500000,700000,600000),
  Temperature=c(75,68,80),
  Elevation=c(1000,800,1200)
)
print(data)

```

	City	Population	Temperature	Elevation
1	City A	5e+05	75	1000
2	City B	7e+05	68	800
3	City C	6e+05	80	1200

13d. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

```

library(shiny)

ui<-fluidPage(
  plotOutput("scatter"),
  tableOutput("table")
)

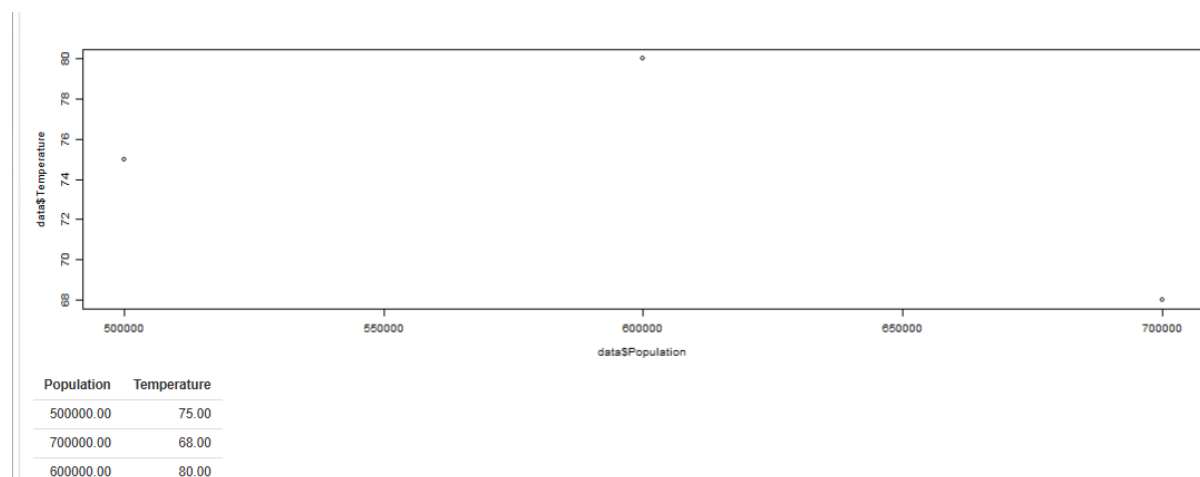
server<-function(input,output){
  data<-data.frame(
    Population=c(500000,700000,600000),
    Temperature=c(75,68,80)
  )
}

```

```

)
output$scatter<-renderPlot({
plot(data$Population,data$Temperature)
})
output$table<-renderTable({
data
})
}
shinyApp(ui,server)

```



14. Dataset: Survey Results

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

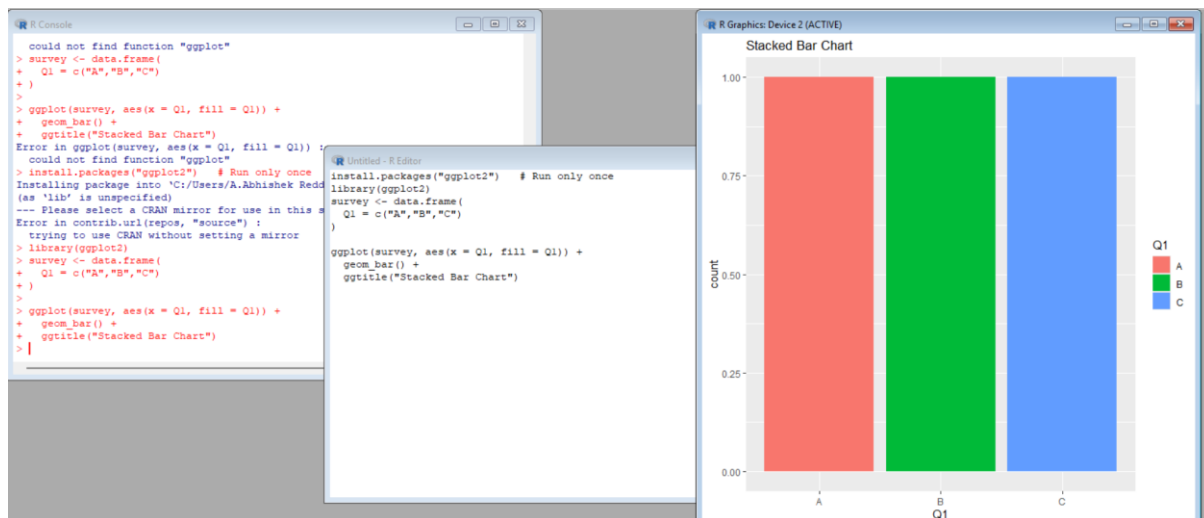
14a.Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

```

install.packages("ggplot2")
library(ggplot2)
survey <- data.frame(
  Q1 = c("A","B","C")
)

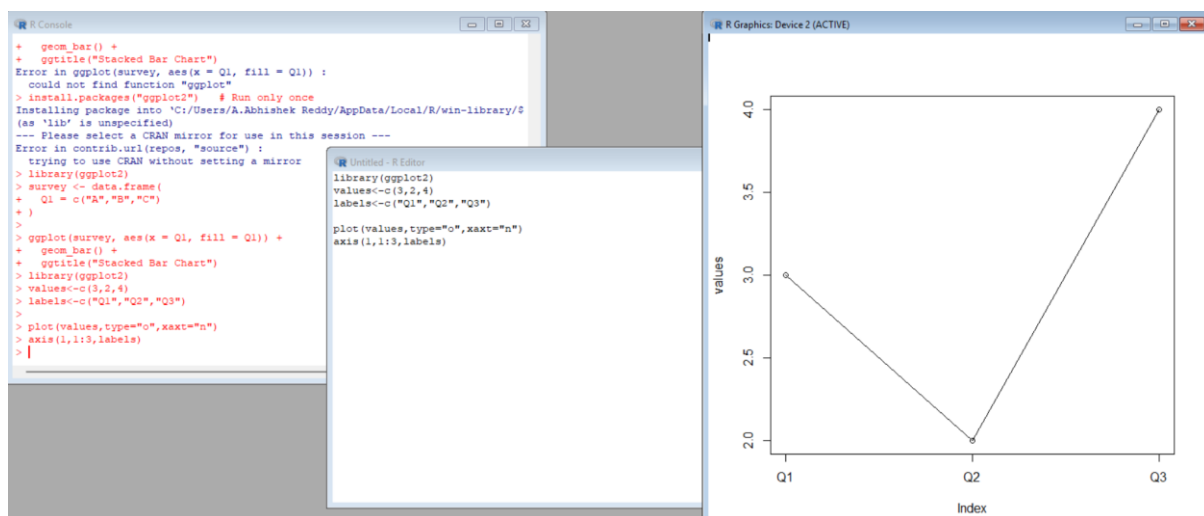
```

```
ggplot(survey, aes(x = Q1, fill = Q1)) +
  geom_bar() +
  ggtitle("Stacked Bar Chart")
```



14b. Generate a radar chart to visualize the overall pattern of responses across all three questions.

```
library(ggplot2)
values<-c(3,2,4)
labels<-c("Q1","Q2","Q3")
plot(values,type="o",xaxt="n")
axis(1,1:3,labels)
```



14c. Build a table to show the survey response data for each respondent. Label the table elements.

```
library(ggplot2)
```

```

survey<-data.frame(
  Respondent=1:3,
  Q1=c("A","B","C"),
  Q2=c("B","A","A"),
  Q3=c("C","D","B")
)
print(survey)

```

```

  Respondent Q1 Q2 Q3
1          1  A  B  C
2          2  B  A  D
3          3  C  A  B

```

14d. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

```

library(shiny)

survey <- data.frame(
  Respondent = c(1,2,3),
  Q1 = c("A","B","C"),
  Q2 = c("B","A","A"),
  Q3 = c("C","D","B")
)

ui <- fluidPage(
  h2("Survey Responses Dashboard"),
  plotOutput("bar"),
  plotOutput("radar"),
  tableOutput("table")
)

server <- function(input, output){
  output$bar <- renderPlot({
    barplot(table(survey$Q1),
             col=c("red","blue","green"),
             main="Stacked Bar Chart")
  })
}

```

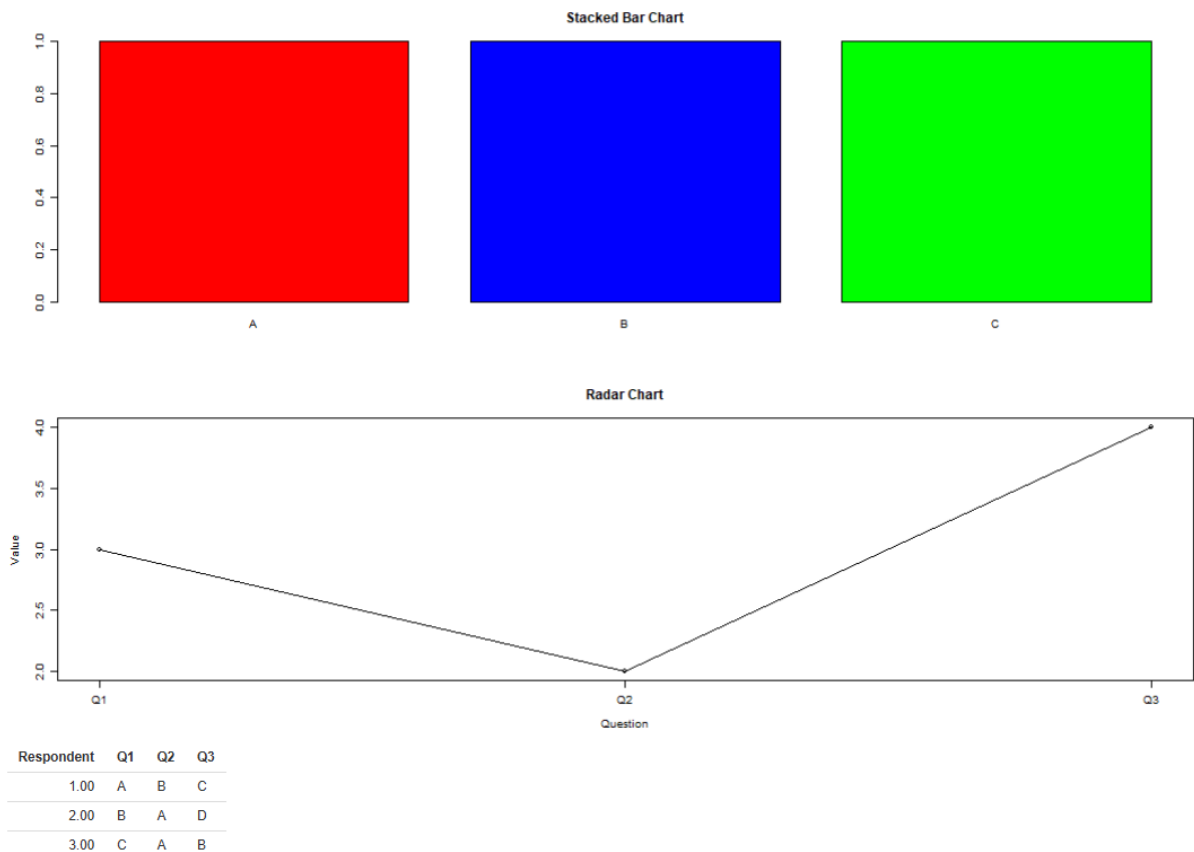
```

output$radar <- renderPlot({
  values <- c(3,2,4)
  plot(values,
    type="o",
    xaxt="n",
    xlab="Question",
    ylab="Value",
    main="Radar Chart")
  axis(1, at=1:3, labels=c("Q1","Q2","Q3"))
})
output$table <- renderTable({
  survey
})
}

```

```
shinyApp(ui, server)
```

Survey Responses Dashboard



Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

15a. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

```
library(ggplot2)
```

```
data<-data.frame(
```

```
Month=c("Jan","Jan","Jan","Feb","Feb","Feb","Mar","Mar","Mar"),
```

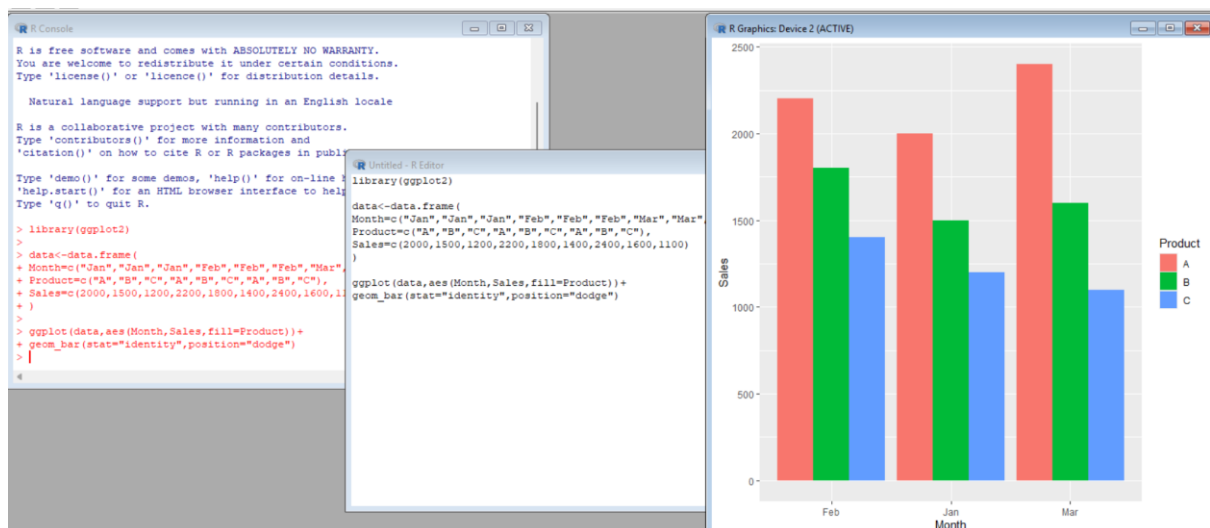
```
Product=c("A","B","C","A","B","C","A","B","C"),
```

```
Sales=c(2000,1500,1200,2200,1800,1400,2400,1600,1100)
```

```
)
```

```
ggplot(data,aes(Month,Sales,fill=Product))+
```

```
geom_bar(stat="identity",position="dodge")
```

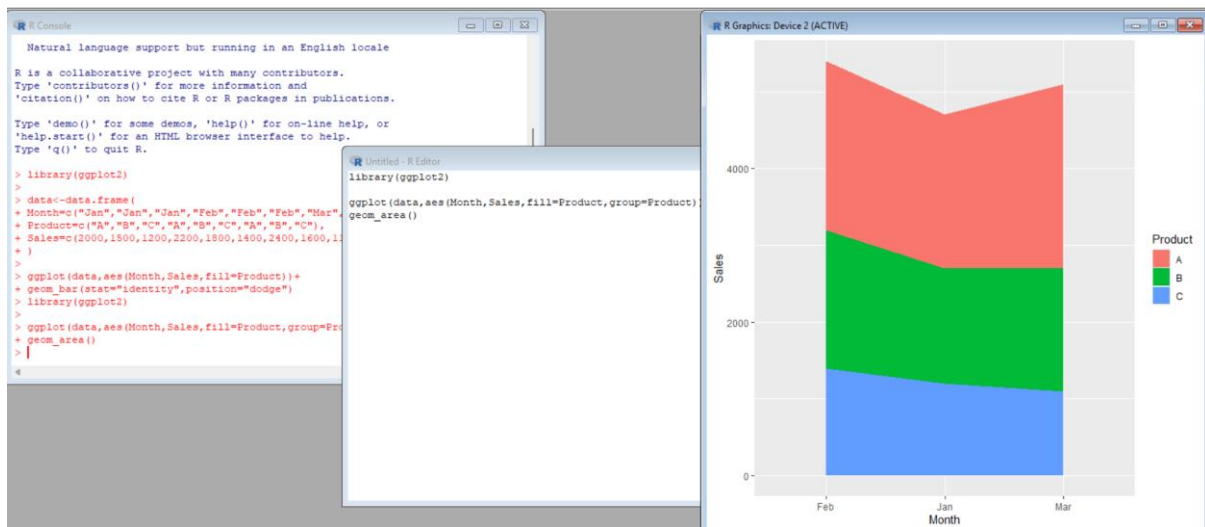


15b. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
library(ggplot2)
```

```
ggplot(data,aes(Month,Sales,fill=Product,group=Product))+
```

geom_area()



15c. Build a table to show the monthly sales data for each product. Label the table elements.

```
library(ggplot2)
```

```
sales<-data.frame(
```

```
Product=c("Product A","Product B","Product C"),
```

```
January=c(2000,1500,1200),
```

```
February=c(2200,1800,1400),
```

```
March=c(2400,1600,1100)
```

```
)
```

```
print(sales)
```

```
      Product January February March
1 Product A    2000     2200   2400
2 Product B    1500     1800   1600
3 Product C    1200     1400   1100
```

15d. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

```
library(shiny)
```

```
library(ggplot2)
```

```
sales <- data.frame(
```

```
Month = c("Jan","Jan","Jan","Feb","Feb","Feb","Mar","Mar","Mar"),
```

```
Product = c("Product A","Product B","Product C",
```

```
"Product A","Product B","Product C",
```

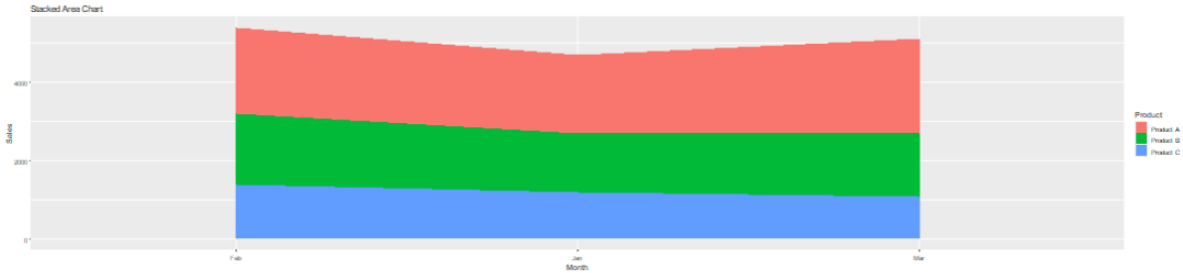
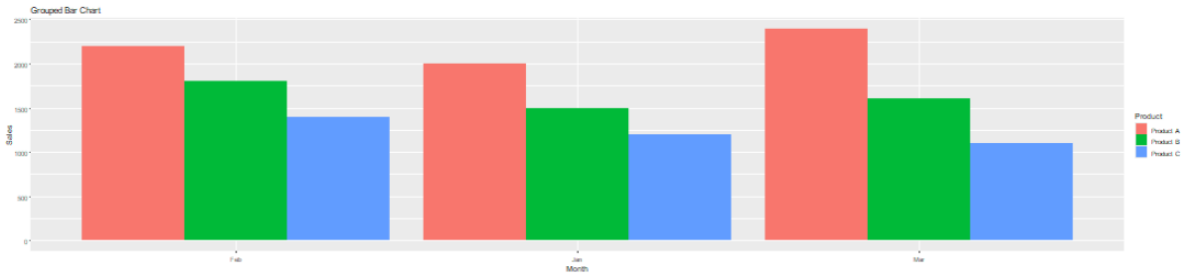
```
"Product A","Product B","Product C"),
```

```

Sales = c(2000,1500,1200,2200,1800,1400,2400,1600,1100)
)
ui <- fluidPage(
  h2("Product Sales Dashboard"),
  plotOutput("bar"),
  plotOutput("area"),
  tableOutput("table")
server <- function(input, output){
  output$bar <- renderPlot({
    ggplot(sales, aes(Month, Sales, fill=Product)) +
      geom_bar(stat="identity", position="dodge") +
      ggtitle("Grouped Bar Chart")
  })
  output$area <- renderPlot({
    ggplot(sales, aes(Month, Sales, fill=Product, group=Product)) +
      geom_area() +
      ggtitle("Stacked Area Chart")
  })
  outut$stable <- renderTable({
    sales
  })
}
shinyApp(ui, server)

```


Product Sales Dashboard



Month	Product	Sales
Jan	Product A	2000.00
Jan	Product B	1500.00
Jan	Product C	1200.00
Feb	Product A	2200.00
Feb	Product B	1800.00
Feb	Product C	1400.00
Mar	Product A	2400.00
Mar	Product B	1600.00
Mar	Product C	1100.00